True frequency cuts at two drive phases

$\varphi = 0$ $\varphi = \pi/2$

Extended static reference; $V/t_0 = 0.5$

Critical static reference; $V/t_0 = 2$

Localized static reference; $V/t_0 = 4$

$\overline{\mathrm{IPR}}$

$\overline{D_n}$

$\overline{D}_{\mathrm{outer}}$

$\Delta\overline{D_n}$

static limit

Dimensionless ω/t_0

Phase π/2 is computed separately because it is absent from the 50-point periodic phase grid.
Dots are computed points; connecting lines guide the eye. Open ω=0 markers are separate frozen static values, not a Floquet limit.